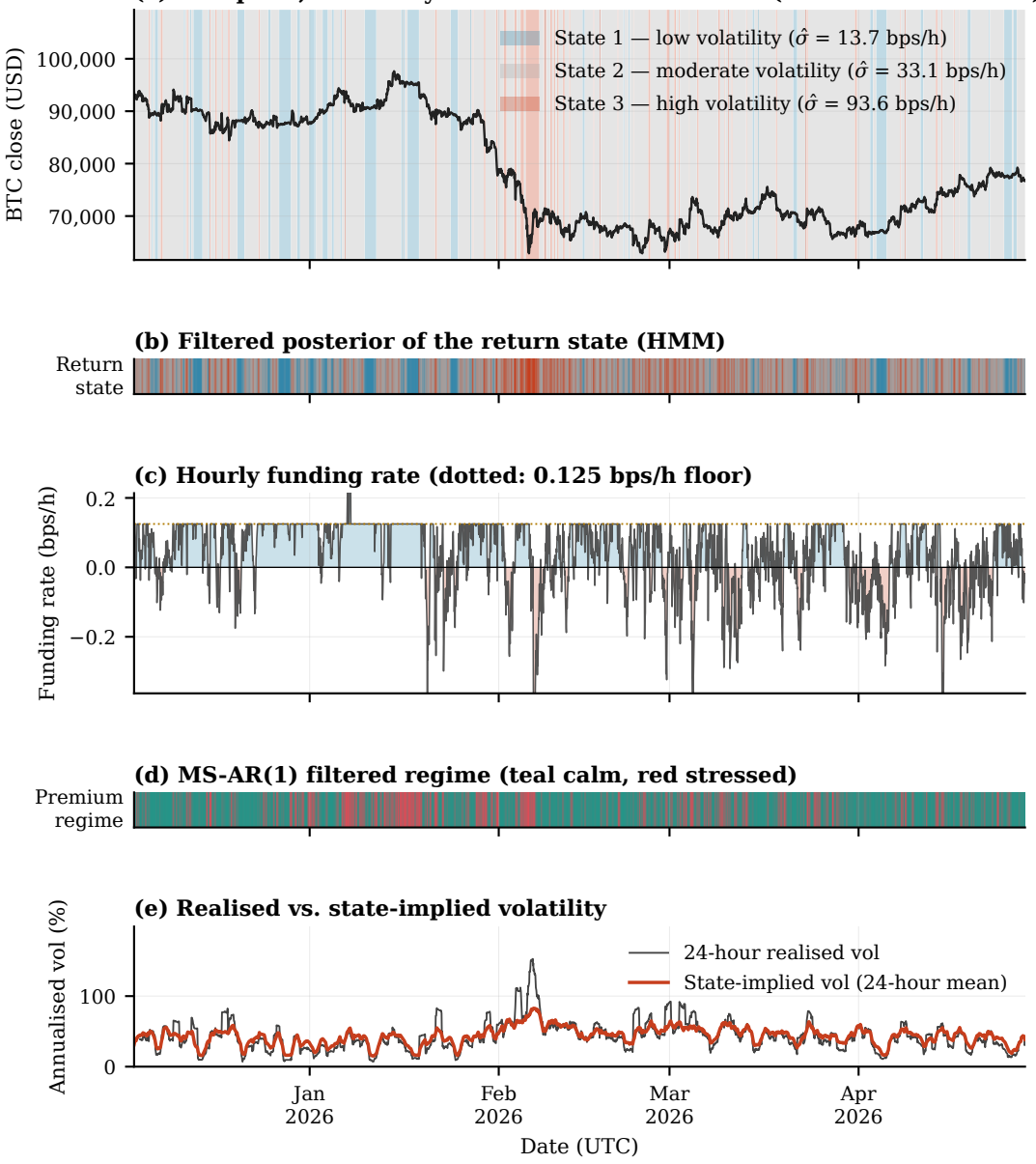

(a) BTC price, shaded by the dominant return state (7-hour median filter)
BTC close (USD)
100,000
90,000
80,000
70,000
State 1 — low volatility ($\hat{\sigma}$ = 13.7 bps/h)
State 2 — moderate volatility ($\hat{\sigma}$ = 33.1 bps/h)
State 3 — high volatility ($\hat{\sigma}$ = 93.6 bps/h)
(b) Filtered posterior of the return state (HMM)
Return state
(c) Hourly funding rate (dotted: 0.125 bps/h floor)
Funding rate (bps/h)
0.2
0.0
−0.2
(d) MS-AR(1) filtered regime (teal calm, red stressed)
Premium regime
(e) Realised vs. state-implied volatility
Annualised vol (%)
100
0
24-hour realised vol
State-implied vol (24-hour mean)
Jan 2026
Feb 2026
Mar 2026
Apr 2026
Date (UTC)